# Doze Detector

"A sleep apnea monitor: your sleep, now under surveillance!"

Ishaan Mehrotra and Shubhika Munot

# Motivation




- Sleep disorders are characterized by repeated interruptions in breathing during sleep, with five or more non-breathing events indicating the condition.[1]
- Affects approximately 25% of men and nearly 10% of women.[2]
- Includes obstructive sleep apnea (OSA), central sleep apnea (CSA), and complex sleep apnea syndrome.[3]
- Symptoms include loud snoring, choking, or gasping during sleep, and excessive daytime sleepiness.[2]
- Existing sleep apnea monitors are limited by delayed response times and incomplete positional data capture.[4]

# Block Diagram

**Input Device:**
Nasal Cannula - Connects to patient and pressure sensor

**Sensors:**
Differential Pressure Sensor - Measures the pressure of the ambient air and inhalation/exhalation.

**Signal Processing 1:**
Differential Amplifier: amplifies the difference between the two inputs (inhalation/exhalation + ambient air)
High Pass Filter: Blocks DC bias from the amplified signal

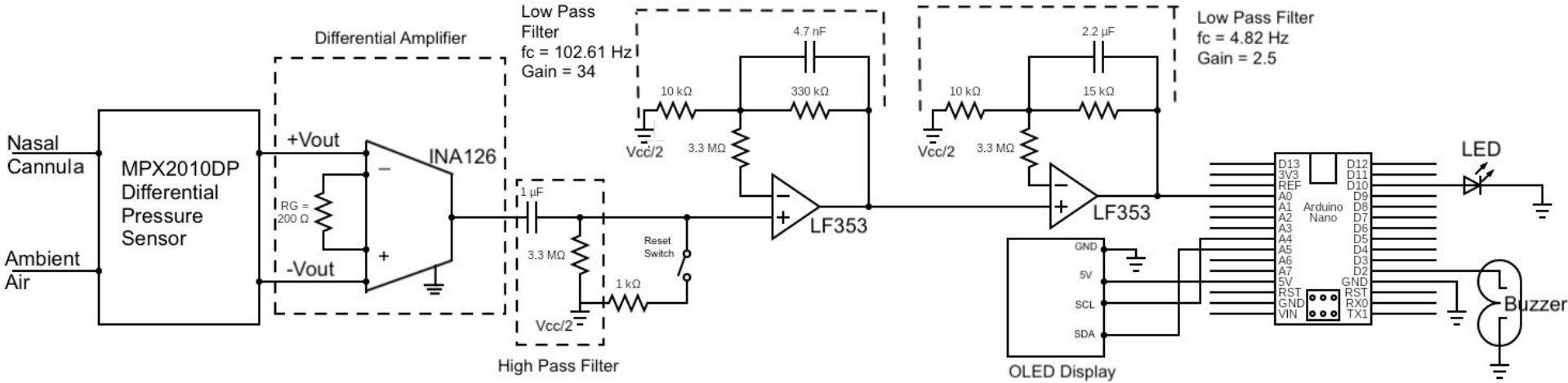
**Signal Processing 2:**
Pressure Amplifier:
Amplifies the pressure signal. Gain set to approximately 34 to sufficiently amplify pressure signal without distorting the signal.

**Signal Processing 3:**
Pressure Sensor Output Filtering:
Type: Low-Pass Filter
Order: First-Order filter.
Cutoff Frequencies: Cutoff frequency round 4.8 Hz. To capture low frequency components of pressure signal related to breathing

**Microcontroller:**
Arduino Nano Every
Convert the analog signals from the filters into digital signals for processing.
Runs analysis algorithm to determine breathing status, apnea events and severity levels

**Display and Alert System:**
OLED Display - Show apnea events and alerts.
LED - Flashes for all apnea severity levels
Buzzer - Buzzes for moderate (lower freq) and severe (higher freq) apnea severity levels

# Final Schematic



# Device Specifications & Testing - Functional

**Pressure Sensor Accuracy and Range:**
The device measures differential pressure from the breathing cycle using an analog pressure sensor.

- Detection Range: 0.0 volts to 5.0 volts.
- Threshold for Detection: Based on noise analysis - ±0.12 V from baseline → **passes; can detect inhale, exhale, no breath**
- Baseline calibration at beginning (average 100 readings) → **passes; baseline set around 2.5 + 20% V**

**Apnea Event Detection:**
Minimum Duration for Apnea Event: The device detects apnea events lasting ≥2 seconds. → **passes; @ 2 seconds of no breath, OLED displays "APNEA"**

Event Detection Rate: Capable of detecting an unlimited number of apnea events per minute. → **device always displays "APNEA" at each no breath event**

**Test Duration:**
The device runs a 3-minute simulation (scaled down equivalent to 8-12 test) → **passes; test goes on and completes after 3 minutes**

Time is divided into 1-minute intervals for tracking the number of apnea events. → **passes; interval reset every minute**

**Severity Classification:**
Events per minute thresholds:

Normal: <2 events/min.
Mild Apnea: 2–4 events/min.
Moderate Apnea: 4–6 events/min.
Severe Apnea: >6 events/min.

# Device Specifications & Testing - Display & Alert

**OLED Display:**

- Time remaining in the test
- Current breathing state: Inhaling, Exhaling, or No Breath.
- Apnea event status, including duration of ongoing apnea events.
- Total number of apnea events detected.
- Number of apnea events in current interval
- Current severity classification (Normal, Mild, Moderate, Severe).

**→ passes all tests and displays the appropriate information on the OLED**

**Alerts:**

LED: Flashes every 250 ms during all apnea classification. **→ passes; starts at mild, continues throughout moderate and severe apnea cases**

Buzzer:

- Frequency of 4000 Hz for severe apnea.
- Frequency of 3000 Hz for moderate apnea.
- Buzzer duration: 100 ms.

**→ passes; buzzes at different frequencies based on severity and no buzzing for mild cases (only light)**

# Test Cases - Primary Function

## 1. Gain

- **Differential Amplifier** (expected gain = **405**):
  Supplied: 4 mVpp
  Output: 1600 mVpp
  Calculated Gain: **400**

- **Filtering Stages** (expected gain = **34*2.5 = 85** ):
  Supplied: 4 mVpp
  Output: 320 mVpp
  Calculated Gain: **80**

## 2. Calibration

- Average out 100 readings from the nasal cannula to calibrate the baseline voltage detection

  → **2.5 ± 20% V**

## 3. Breathing

- Above baseline + threshold shows **exhaling** (air coming out of nose), while below baseline - threshold shows **inhaling** (air coming into nose)
- No air movement through nasal cannula shows **no breath**

# Test Cases - Primary Function

4. **Severity Classification**

**an event is classified as no breath for >2 seconds**

- 1 event per minute (Normal)

  → no flashing light, no buzzer sound

- 2-3 events per minute (Mild)

  → only flashing light, no buzzer sound

- 4-5 events per minute (Moderate)

  → flashing light, low frequency buzzer sound

- 6 or more events per minute (Severe)

  → flashing light, high frequency buzzer sound

# Test Cases - Environmental

**5. Environmental Effects**

We tested the effects of fan distance and fan speed on the functionality of our device

We tested:

**False positive errors** by breathing for 2 mins to see if any apnea events are detected

**False negative errors** by not breathing for 2 mins to see if any breathing events were detected

We performed a **Chi-square Goodness of Fit Test!**

To see if observed distribution of errors (false positives/false negatives) across different fan distances/speeds differs significantly from a hypothesized uniform distribution (i.e., no difference in error rates across distances/speeds)

# Test Cases - Fan Distance

**False Positive:**
Null Hypothesis ($H_0$): The false positive errors are the same across all fan distances.

Alternative Hypothesis ($H_a$): The false positive errors are not the same across all fan distances. (i.e. fan distance affects false positive errors)

Results:
$X^2 (= 3.66) < \text{Critical Value}_{3,0.05} (= 7.815)$

Cannot reject the null hypothesis.

**∴ The fan distance does not affect false positive errors.**

**False Negative:**
Null Hypothesis ($H_0$): The false negatives errors are the same across all fan distances.

Alternative Hypothesis ($H_a$): The false negative errors are not the same across all fan distances. (i.e. fan distance affects false negative errors)

Results:
$X^2 (= 33.62) > \text{Critical Value}_{3,0.05} (=7.815)$

Reject the null hypothesis.

**∴ The fan distance significantly affect false negative errors.**

# Test Cases - Fan Speed

**False Positive:**
Null Hypothesis ($H_0$): The false positive errors are the same across all fan speeds.

Alternative Hypothesis ($H_a$): The false positive errors are not the same across all fan speeds. (i.e. fan speed affects false positive errors)

Results:
$X^2 (= 3) < \text{Critical Value}_{3,0.05} (= 7.815)$

Cannot reject the null hypothesis.

**∴ The fan speed does not affect false positive errors.**

**False Negative:**
Null Hypothesis ($H_0$): The false negatives errors are the same across all fan speeds.

Alternative Hypothesis ($H_a$): The false negative errors are not the same across all fan speeds. (i.e. fan speed affects false negative errors)

Results:
$X^2 (= 35.3) > \text{Critical Value}_{3,0.05} (= 7.815)$

Reject the null hypothesis.

**∴ The fan speed significantly affect false negative errors.**

# Test Cases - Environmental

| **Correct Detection Test** | *Breathing for 2 min and stopping every 30 seconds for 5 seconds* | |
|---|---|---|
| **Condition** | **Average Apnea Event Detected (out of 3)** | **Average Time Delay for Apnea Detection (seconds)** |
| Fan Distance | | |
| No Fan | 3 | 0 (instant) |
| 1 foot | 2.67 | 2.67 |
| 2.5 feet | 3 | 1.33 |
| 5 feet | 3 | 0 (instant) |
| Fan Speeds (@ 2.5 feet) | | |
| Low | 3 | 2.33 |
| Medium | 3 | 1.33 |
| High | 3 | 4.67 |

# Test Cases - Human

## 6. Motion Effects

| Condition | False Positive | False Negative |
|---|---|---|
| Head side-to-side | 0 | 0 |

| Condition | Average Apnea Event Detected (out of 3) | Average Time Delay for Apnea Detection (seconds) |
|---|---|---|
| Head side-to-side | 3 | 2 |

**We tested the calibrated baseline under all test cases:
$2.5 \pm 20\%$ V




# Clinical Comparison with Existing Devices

- **ResMed ApneaLink Air** [5]

  → records respiratory effort, pulse, oxygen saturation, and nasal flow

  → uses respiratory sensors & pulse oximetry

  → provides Apnea-Hypopnea (AIH) scores

  → worn around the chest with nasal cannula

  → lightweight (2.5 oz), compact & portable

  → single-night snapshot only, may require repeat testing, no real-time monitoring capability

- **WatchPAT ONE** [6]

  → measures oxygen saturation, heart rate, & actigraphy

  → uses Peripheral Arterial Tone (PAT) tech with finger-based plethysmography to specifically diagnose obstructive sleep apnea

  → detects respiratory changes through vascular changes

  → single-use disposable, minimal attachments, higher cost, battery dependent, internet connection needed for data

# Limitations & Future Improvements

## Limitations

1. **Reduced Accuracy due to Noisy Environments**

   → external disturbances lead to **false negatives**

2. **Single Pressure Sensor Dependency**

   → does not account for bilateral nasal breathing or breathing through the mouth at night

3. **Uncomfortable Design for Patients**

   → device setup currently is not ergonomic enough to safely wear all night for detection

## Improvements

1. **Improve Robustness**

   → incorporate noise calculations to prevent fluctuations (↓ **false negatives** )

   → also test for coughing, speaking, etc.

2. **Dual-Sensor System**

   → add second sensor to monitor mouth breathing to improve accuracy

3. **Increase Wearability & User Friendliness**

   → make device more compact to improve user comfort during sleep and connect to smartphone to improve usability

# Thank you!

# References

[1] T. Young et al., "Sleep Disordered Breathing and Mortality: Eighteen-Year Follow-up of the Wisconsin Sleep Cohort," Sleep, vol. 31, no. 8, pp. 1071-1078, Aug. 2008.
[2] T. Young, M. Palta, J. Dempsey, J. Skatrud, S. Weber, and S. Badr, "The occurrence of sleep-disordered breathing among middle-aged adults," New England Journal of Medicine, vol. 328, no. 17, pp. 1230-1235, 1993
[3] A. Malhotra and D. P. White, "Obstructive sleep apnoea," The Lancet, vol. 360, no. 9328, pp. 237-245, Jul. 2002, doi: 10.1016/S0140-6736(02)09464-3.
[4] S. K. Longmore, G. Y. Lui, G. Naik, P. P. Breen, B. Jalaludin, and G. D. Gargiulo, "A Comparison of Reflective Photoplethysmography for Detection of Heart Rate, Blood Oxygen Saturation, and Respiration Rate at Various Anatomical Locations," Sensors, vol. 19, no. 8, p. 1874, Apr. 2019, doi: 10.3390/s19081874.
[5] ResMed, "ApneaLink Air - Home Sleep Testing Device," ResMed.com, 2015. [Online]. Available: https://www.resmed.com/en-us/healthcare-professional/products/diagnostics/apnealink-air/
[6] I. Pillar and A. Berall, "Advances and innovations in the field of home sleep medicine," Canadian Journal of Respiratory Therapy, vol. 56, pp. 52-56, 2020, doi: 10.29390/cjrt-2020-015.

# Appendix

**Fan Distance:**

*False Positives (breathing for 2 mins straight to see if any apnea events are detected)*

Raw Data:

| Fan Distance | Trial 1 | Trial 2 | Trial 3 | Sum (Observed) |
|---|---|---|---|---|
| No fan | 0 | 0 | 0 | 0 |
| 1 feet | 1 | 0 | 1 | 2 |
| 2.5 feet | 0 | 0 | 1 | 1 |
| 5 feet | 0 | 0 | 0 | 0 |
| Total | | | | 3 |

Chi-square statistic calculations:

$$\text{Expected Count per Distance} = \frac{\text{Total Observed}}{\text{Number of Categories}} = \frac{3}{4} = 0.75$$

$$\chi^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$

| Fan Distance | Observed ($O_i$) | Expected ($E_i$) | $(O_i - E_i)^2 / E_i$ |
|---|---|---|---|
| No fan | 0 | 0.75 | 0.75 |
| 1 feet | 2 | 0.75 | 2.08 |
| 2.5 feet | 1 | 0.75 | 0.08 |
| 5 feet | 0 | 0.75 | 0.75 |
| Chi-square statistic ($\chi^2$) | | | 3.66 |

Using the Chi-square distribution table for 3 degrees of freedom and a significance level of 0.05 (α = 0.05):
Critical value = 7.815

# Appendix

**Fan Distance:**

*False Negatives (not breathing for 2 mins to see if any breathing events were detected)*

Raw Data:

| Fan Distance | Trial 1 | Trial 2 | Trial 3 | Sum (Observed) |
|---|---|---|---|---|
| No fan | 0 | 0 | 0 | 0 |
| 1 feet | 8 | 9 | 6 | 23 |
| 2.5 feet | 6 | 5 | 5 | 16 |
| 5 feet | 2 | 0 | 1 | 3 |
| Total | | | | 42 |

Chi-square statistic calculations:

$$\text{Expected Count per Distance} = \frac{\text{Total Observed}}{\text{Number of Categories}} = \frac{42}{4} = 10.5$$

$$\chi^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$

| Fan Distance | Observed ($O_i$) | Expected ($E_i$) | $(O_i - E_i)^2 / E_i$ |
|---|---|---|---|
| No fan | 0 | 10.50 | 10.50 |
| 1 feet | 23 | 10.50 | 14.88 |
| 2.5 feet | 16 | 10.50 | 2.88 |
| 5 feet | 3 | 10.50 | 5.36 |
| Chi-square statistic ($\chi^2$) | | | 33.62 |

Using the Chi-square distribution table for 3 degrees of freedom and a significance level of 0.05 ($\alpha = 0.05$):
Critical value = 7.815

# Appendix

**Fan Speed:**

*False Positives (breathing for 2 mins straight to see if any apnea events are detected)*

Raw Data:

| Fan Speed | Trial 1 | Trial 2 | Trial 3 | Sum (Observed) |
|---|---|---|---|---|
| No fan | 0 | 0 | 0 | 0 |
| Low | 0 | 0 | 0 | 0 |
| Medium | 0 | 0 | 1 | 1 |
| High | 0 | 0 | 0 | 0 |
| Total | | | | 1 |

Chi-square statistic calculations:

$$\text{Expected Count per Distance} = \frac{\text{Total Observed}}{\text{Number of Categories}} = \frac{1}{4} = 0.25$$

$$\chi^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$

| Fan Speed | Observed ($O_i$) | Expected ($E_i$) | $(O_i - E_i)^2 / E_i$ |
|---|---|---|---|
| No fan | 0 | 0.25 | 0.25 |
| Low | 0 | 0.25 | 0.25 |
| Medium | 1 | 0.25 | 2.25 |
| High | 0 | 0.25 | 0.25 |
| Chi-square statistic ($\chi^2$) | | | 3 |

Using the Chi-square distribution table for 3 degrees of freedom and a significance level of 0.05 ($\alpha = 0.05$):
Critical value = 7.815

# Appendix

**Fan Speed:**

*False Negatives (not breathing for 2 mins to see if any breathing events were detected)*

Raw Data:

| Fan Speed | Trial 1 | Trial 2 | Trial 3 | Sum (Observed) |
|---|---|---|---|---|
| No fan | 0 | 0 | 0 | 0 |
| Low | 10 | 11 | 12 | 33 |
| Medium | 6 | 5 | 5 | 16 |
| High | 10 | 11 | 10 | 31 |
| Total | | | | 80 |

Chi-square statistic calculations:

$$\text{Expected Count per Distance} = \frac{\text{Total Observed}}{\text{Number of Categories}} = \frac{80}{4} = 20$$

$$\chi^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$

| Fan Speed | Observed ($O_i$) | Expected ($E_i$) | $(O_i - E_i)^2 / E_i$ |
|---|---|---|---|
| No fan | 0 | 20 | 20 |
| Low | 33 | 20 | 8.45 |
| Medium | 16 | 20 | 0.8 |
| High | 31 | 20 | 6.05 |
| Chi-square statistic ($\chi^2$) | | | 35.3 |

Using the Chi-square distribution table for 3 degrees of freedom and a significance level of 0.05 ($\alpha = 0.05$):
Critical value = 7.815